

Practice Quiz: Planning

Part A: Multiple Choice

1. A multi-step policy specifies: A) A single action B) A sequence of actions over T future timesteps C) A probability distribution over actions D) A belief update rule
2. The total EFE of a 3-step policy is computed by: A) Taking the EFE at the first step only B) Taking the maximum EFE across steps C) Summing $G_{\tau}(\pi)$ for $\tau = 0, 1, 2$ D) Multiplying the three step EFEs
3. Marginal Message Passing differs from single-step inference because it: A) Uses the C-vector B) Infers beliefs at multiple time points simultaneously C) Only works for gridworlds D) Does not use the A-matrix
4. For a 3-action system with temporal depth $T = 4$, the number of exhaustive policies is: A) 12 B) 64 C) 81 D) 256
5. In a gridworld, walls are encoded by: A) Removing states from the state space B) Setting the B-matrix to have self-transitions at wall positions C) Modifying the A-matrix D) Setting C to zero at wall states
6. Why does the T-maze require $T \geq 2$ for reliable performance? A) The environment has more than 2 states B) The agent must visit the cue (step 1) before choosing an arm (step 2) C) The B-matrix requires 2 transitions D) Precision γ must equal 2
7. `plot_simulation_dashboard()` displays: A) A single belief plot B) 5 panels: beliefs, VFE, predictions, prediction errors, and EFE C) The gridworld state space D) The A and B matrices

Part B: Short Answer

1. Write code to define all 2-step policies for a 3-action system. How many policies are there? Display them as a list of lists.

2. Explain why the computational cost of planning scales exponentially with temporal depth T . Describe one strategy to manage this complexity (e.g., pruning, hierarchical policies, or Monte Carlo sampling).
3. Design a delayed-reward gridworld where a greedy ($T = 1$) agent gets stuck but a planning ($T = 3$) agent succeeds. Sketch the grid, identify the obstacle, and explain why depth matters.